ILLINOIS TECH
Stuart School of Business

ILLINOIS TECH | Stuart School of Business

Instructions:

1. Make sure that you have carefully read and fully understood the question before answering it. Answer the question fully but concisely and as directly as possible.

2. Answer the question in your own words. Do not copy any text from the readings or other sources. **The assignment must be your own work.**

In this final assignment you will create a composite roadmap for strategically transforming an organization using (AI) technologies. In each of the four assignments leading up to the final composite roadmap deliverable, you completed activities related to this composite roadmap's development. That work informed your thinking for the completion of this final roadmap. A high-level overview of the structure for the roadmap is shown below.

In assignment #5, you will be working on building the "**Composite Roadmap**."

Overall Roadmap:

1. Executive summary
    - *Ideas: Context, Actions Envisioned, Results Sought*
    - *Ideas: Leadership and digital technologies assessment*
2. Current state
    - *Ideas: Strategy today*
    - *Ideas: Is technology aligned with strategy*
        - *Ideas: Are there different or better strategies to use going forward?*
        - *Ideas: How could AI fit with the chosen strategy?*
        - *Ideas: Could your use of (AI) tech bring new value to the organization and how?*
3. Proposed initiative
    - *Ideas: Identify three problems/opportunities/situations*
        - *Ideas: Vision/mission/goal alignment*
        - *Ideas: External/Internal Environment*
        - *Ideas: SWOT analysis*
4. Plans of action and criteria for success
    - *Ideas: Build*

ILLINOIS TECH | Stuart School of Business

- *Ideas: Deploy*
- *Ideas: Manage/Measure*
- *Ideas: Timeline or Steps*

Backdrop and <u>Questions to answer</u> about the Composite Roadmap:

Consider a working environment. It could be your current organization, an organization where you have previously worked, or a fictional organization.

Write a narrative report of **up to 4,000 words,** detailing the *composite roadmap* for your chosen organization. Use the following questions to guide your thinking:

Q1:
Did you craft a compelling Executive Summary?

Q2:
What is your best assessment of the Current State of the organization that you have chosen to strategically transform with (AI) technology?

Q3:
Have you articulated your Proposed Initiative clearly?

Q4:
Is there a clear description of your Plan of Action and Criteria for Success?

Q5:
Do the sections of your roadmap integrate well and are coherent?

Q6:
Is the Structure and Logic of your Writing clear?

Note:

You may include any of your previously submitted assignments exactly as you have written them. If you use citations, please include a References section at the end of the document and use APA style. Ensure that your writing is coherent and clear.

Start writing below this line:

ILLINOIS TECH
Stuart School of Business

ILLINOIS TECH | Stuart School of Business

Rudra More
BUS 510 -S02

Executive Summary

Project Overview:

Goal:

The objectives of the plan are as follows:

• Use drone technology integrated with artificial intelligence to optimize delivery times and involved processes in the next fiscal quarter

• Improve package maintenance and quality control of all items shipped through computer vision.

• Enhance the security of package deliveries by leveraging advanced AI and tracking technologies

Scope of the Project: This project will specifically focus on the integration of AI and drone technology in the USPS's last-mile delivery process, particularly targeting rural and remote regions where traditional delivery methods face logistical challenges. The project will impact the delivery operations, logistics management, and customer service support departments by introducing AI-powered tracking systems and real-time monitoring tools. It will cover improvements in package sorting, route optimization, and delivery verification. However, the project will not include changes to USPS's long-haul delivery operations or internal processing networks outside of the last-mile service. It is limited to enhancing the efficiency of package delivery, security of goods, and customer experience within the final stages of the delivery process. Integration with current systems, employee training, and the testing of new AI-driven tools in selected regions will also fall within the project's scope.

Strategic Alignment:

Supports USPS's goal of faster, reliable, and efficient services: The use of AI-powered drones aligns with USPS's goal to improve delivery speed and efficiency, particularly in remote areas, enhancing customer satisfaction and service reliability.

Aligns with USPS's vision of a digitally enabled, future-ready logistics platform: Implementing AI and drone technology supports USPS's strategy to modernize its logistics infrastructure and adapt to the growing demands of e-commerce, furthering its transformation into a future-ready organization.

Contributes to cost optimization and financial sustainability: Drone deployment reduces reliance on traditional delivery methods and workforce, lowering operational costs while ensuring USPS

ILLINOIS TECH | Stuart School of Business

meets its universal service obligation. This aligns with USPS's goals for innovation, cost reduction, and long-term financial stability

The USPS has key objectives in place as part of the five-year plan project to improve delivery processes and showcase a strong workforce that wants to maintain the highest customer satisfaction. Within the objectives is the goal to create a more technology integrated infrastructure to ensure better communication with customers as well as improve brand awareness by showcasing their services to far-to-reach areas. Providing costs at a more acceptable rate and sustainable methods of delivery packaging, the organization would be able to put themselves in a position of financial stability through less reliance on the workforce proving that they are moving in the right direction of the USPS mission.

Key Benefits and Expected Outcomes:

Value Proposition: With the implementation of the above approaches, there would be a marginal decrease in delivery times through the avoidance of public transport and traffic for heavily populated areas that experience rush hour.

Expected Impact: Integrating such technology will require several tests that will be put in place to observe successful deliveries in areas close to central hubs over the short term while regular delivery services will proceed over long distances. This would ultimately lead to improving overall customer confidence in the brand due to the reliable delivery rates and package handling in the long run.

High-Level Implementation Plan:

Phases and Milestones:
Each phase of drone implementation will be divided into groups to measure the success rate in each category.

Phase 1: Pilot Testing Phase 1 will include Drone fleets will be programmed by drone pilots for each zip code over a short period of time to evaluate the feasibility under a given set of conditions and evaluate feedback of customers based on their experience.

Phase 2: Infrastructure Integration Phase 2 will include the deployment of drone logistics and software components through its integration with the current USPS IT systems to provide drone tracking, package maintenance and delivery status from the mainframe.

Phase 3: Full Deployment Phase 3 will include upgrading the software and server hubs to maintain bigger fleets as well as upgraded drone vehicles to expand delivery to more rural areas as well as increasing the number of packages that can be delivered in the same area with enhanced automotive capabilities.

ILLINOIS TECH | Stuart School of Business

Resources and Budget:

As per the current budget, this project would include &50 million for research and development to integrate unmanned automotive technology in the working infrastructure that will cover all courses and training for drone piloting as well as acquiring drone from heavily researched corporations that have the highest successful drone flights recorded. Along with drone pilots, each area code will be handled by a separate team of technicians and engineers with specialized machine learning and computer vision knowledge for drone maintenance and tracking with a support team handling in flow of data to be analyzed for future flights.

Risk Management:

Potential Challenges:
The main concern for drone delivery is the FAA guidelines which constitute the right to drone flight in certain areas of the US which will be a major hurdle for deployment on a large scale. Additionally, significant changes in the workforce would be seen for delivery drivers who would otherwise be trained as drone pilots who would be required to train in uncertain weather conditions until further geographical data has been gathered.

Contingency Plans:
In case of emergencies, a secondary drone will be stationed on standby with the worst case being to use traditional services. Each drone pilot would undergo regulatory assessments regarding use of drones depending on the area to stay within the guidelines and help promote safe drone flight delivery services for customers who wish to use the service in the future.

Conclusion:

Call to Action & Vision for the Future:
As the head of this project. I urge stakeholders to approve funding for drone delivery pilot testing and support AI integration to enhance USPS operations. Successful implementation will position USPS as a leader in logistics, reduce costs, improve customer satisfaction, and promote sustainability, ensuring USPS remains competitive and efficient in the evolving digital economy.

Current State

Current Organizational Strategy

The USPS or as its full form United States Postal Service has been a major contributor to bringing families together since the days of Abraham Lincoln. Their objective is to be the primary mode of delivering mail to any part of the country for all citizens. As of this year they initiated a plan “Delivering for America", a 10-year strategic plan launched in March 2021 that seeks to address both their financial and operational challenges and break even by the end of 2031 in order to meet their goals pf efficiency, expanded delivery service and their improvement of the brand in the eyes of the people.

Their current strategy as per their 10 year plan objective is to create a commitment to at least sending 6 to 7 mails per day, a modernized fleet that is moving towards more sustainable fuel usage, increasing use of integrated communication with customers regarding routines that run in real time assessing where the packages are at any moment of time, a more supportive legislative, a increasing rationalization of prices aimed at establishing more confidence in the consumer and helping the workforce meet the continued growth through a reduction of pre-career workforce turnover by 50%. While these changes have been taken into a effect, according to their two year report since the start of the implementation of this project, they have seen a slow increaser in profits by about 7- 8% regarding all forms of premium and standard services such as reaching packages, licensed goods, etc. which does not bode well in order to meet their financial goal by the end of the following year.

Cost Leadership Strategy

The USPS has been running a basic cost leadership strategy in the terms of providing different forms of low-cost carrier for different needs of the citizens to compete with upcoming companies of the time such as FedEx, UPS, etc. This has worked towards their disadvantage bringing record loses north of $160 billion in terms of capital funding which is not only taken through taxes but also taking into account increasing costs of equipment for the warehouses along with deliveries to areas with low staff competency. Through their development they have constantly faced pension liabilities and operational costs which have put a strain on consumer demand for the service as well as reliance from legislations to rule in favor of bringing further funding to improve the same.

Recommended Strategy Moving Forward

To ensure long-term financial sustainability, USPS should shift from a cost leadership strategy to a differentiation strategy driven by technological innovation. By focusing on improving service quality through advanced technology, USPS can offer unique value that sets it apart from competitors like FedEx and UPS. This shift would allow USPS to enhance customer satisfaction and attract more customers by offering reliable, fast, and technology-driven services. Incorporating AI and drone technology into its operations would help USPS move beyond price competition, positioning itself as a leader in innovative mail delivery solutions that cater to the evolving needs of both urban and rural communities.

Drone technology is one such method that has been used in variety of applications from the army communications generation to providing unmanned flight a simpler modification. This age has brought artificial intelligence being implemented on almost all devices and with its aid, would reach customers at a more guaranteed rate rather than the acquisition of electric vehicles which is still under research and development towards sustainability. These drones can be logged with the addresses of multiple addresses through a central server routing the easiest path to each customer rather than take the long way. This method would save up funding used to create small amounts of revenue to improving consumer confidence through means of connecting customers

ILLINOIS TECH | Stuart School of Business

directly with the senders through a visual camera which can prove the package has reached its destination at the time allotted.

This can be used across multiple packages large and small as drone helicopters and pods have their own storage facility that allow them to carry multiple packages in the same area or zip coded or heavy packages that would slow down traffic and create further problems for the public. With the growing need of public to send out multiple packages from remote locations as well as heavy focused areas of poverty, this provides a mean of not only increasing wider range of consumers that the organization will be aware of but also drive towards their need for development in right direction. The human aspect that makes Postmasters and Local carriers the direct link to customers would be by a constant professional in contact with the sender and messenger alerting them of the logs of the health of the package as well as the remaining time to its destination. This will not only help bring out more human practices of bonding with customers but also help supervisors better understand their needs. It is the basis of their image that consumers get the best possible services and show that their organization meets such standards from where it started.

AI Fit in the Recommended Strategy

AI technology and unmanned drones perfectly fit into USPS's strategy for modernization and efficiency. AI can optimize delivery routes and enhance package tracking, enabling USPS to provide real-time updates to customers. This not only improves delivery times, particularly in remote or hard-to-reach areas, but also boosts customer trust in the service. Furthermore, drones equipped with AI could lower operational costs by reducing fuel consumption, labour, and vehicle maintenance. AI's ability to streamline operations and enhance security through real-time monitoring would ensure more secure and efficient deliveries, aligning with USPS's strategic goals for improvement in operational efficiency and customer communication.

Use of Strategy to Innovate and Create New Value

Implementing a differentiation strategy through AI and drone technology would open new opportunities for USPS to innovate and add value. Drones equipped with AI can handle multiple deliveries at once, improving efficiency and reducing operational costs. This technology would enable USPS to reach underserved areas faster and more reliably, expanding their customer base. AI can also provide predictive insights into fleet maintenance, reducing downtime and increasing overall efficiency. Additionally, enhanced customer communication through AI would build stronger relationships with customers by providing real-time updates and better package tracking. These innovations not only modernize USPS's services but also strengthen its competitive position and drive growth in an increasingly tech-focused market.

Proposed Initiative

Identifying Problems Solvable by AI

ILLINOIS TECH | Stuart School of Business

The USPS (unites states postal service) has been a major department in mailing packages that has spread its presence both nationally and internationally. While their commitment to improving rate of package delivery has seen minor development, they are still facing concern in three particular categories. The biggest concern at the moment is sending packages successfully and within the time period suggested by the service. This has been combatted with multiple branches upgraded to central hubs in remote areas, this has not shown to improve the confidence in the customer given that most packages are sent days later or never found by employees even after repeated customer service complaints and regular check-ups regarding package condition. Another major concern is the growing defect in the "life-force" of said packages sent to the local branch. Many of the packages such as poultry, sensitive information such as checks or funds that are both time sensitive and require strict level of confidentiality seem to be misplaced more often incurring additional fees to the public if they are not delivered on time. This has been a major problem for nursing practitioners who rely on the service to send important reports and live organs for life threatening surgeries which when not given on time, threatens the life of the patient on the other side of the country. A third case is the monitoring of packages from the time it is received by the USPS to its initial time of delivery. A case has been made that has shown that multiple funds have been reportedly gone missing due to mishandling of the packages or unreported stolen items that have to yet to be recovered.

<u>Strategic Planning and AI Implementation</u>

- *Alignment of Mission/Vision/Goals with AI Solutions*

  The mission of USPS to create rational and reduced pricing of mail services to all citizens. The above concerns require decreasing package delivery time with better monitoring of all packages from the time that the services are acquired to the package's delivery to the intended recipient. One of the goals aim at improving delivery of at least 6 to 7 packages per week which can be reduced significantly through unmanned aerial vehicles that are shown to be more reliable in more dynamic environment. With the integration of artificial intelligence software in the current systems, tracking and monitoring of packages would be an improvement meeting the required goal as well as showing a more developed approach to meet the rise in demand from customers. This would also aid in meeting the required reduced pricing objective as it would require less labour force to transfer the same product faster and efficiently.

- *Assessment of External and Internal Environments*

  While USPS has critical objectives in place to meet the increasing demand, understanding both their internal and external environment will provide a deeper analysis into how they are progressing and whether the objectives are creating a positive response from consumers. The competition has a head start in researching similar areas but lack the funds and reach that USPS have shown to have virtually and physically. Amazon has started putting out beta tests for such unmanned automobiles on the ground to deliver packages from warehouses with significant success which have bolstered the public to use their services to send their own packages while UPS and FedEx have built a reliable customer

base. Internally, USPS have created multiple branches nationwide reducing the complexity of meeting requirements in order to join the workforce but have shown to be stretched too thin due to time taken for packages to reach from the central hub to said branches. This has been shown to have multiple drawbacks given that teens and high schools who have not yet reached the mature age are able to join and while they do provide a stable self-serving role to assist the community, are shown to have raised more concerns by the public due to misplaced or wrongly delivered packages.

- *SWOT Analysis for AI-Related Problems and Opportunities*

Strengths:
USPS has a robust workforce, including 73,000 veterans, delivering about 44% of global mail. They offer cost-effective services compared to competitors and aim to continuously improve operations. The integration of AI and drones, supported by partnerships with companies like DVaerialCAM and Instant Eye Robotics, will significantly enhance package delivery speed and security. Drones will allow USPS to focus more on customer relations by automating routine tasks and ensuring package safety from dispatch to delivery.

Weaknesses:
A major weakness is the strain on finances. Securing funding for AI and drone technologies would likely depend on taxpayer dollars, further depleting resources already impacted by the COVID-19 pandemic. Additionally, retraining the workforce—especially veterans—to adapt to new technologies could be challenging, as they may struggle to quickly acquire the necessary technical skills to meet growing consumer expectations for automated delivery systems.

Opportunities:
Drones present an opportunity for USPS to rebuild consumer confidence by ensuring faster, more secure deliveries. AI integration could also enable personalized customer surveys, collecting valuable data for deep analytics to better understand customer preferences and improve service. These innovations will position USPS as a forward-thinking organization, setting new standards in mail delivery, especially in underserved areas.

Threats:
A significant threat comes from competitors, such as FedEx and UPS, who are also adopting similar technologies. Cybersecurity risks, particularly hacking, could pose serious challenges, leading to the potential for stolen packages or compromised delivery systems. Without strong encryption and security measures, USPS's drone network could be vulnerable, leading to loss of trust and operational inefficiencies.
By addressing weaknesses and mitigating threats, USPS can leverage its strengths and capitalize on emerging opportunities to stay competitive and relevant in the evolving logistics landscape.

ILLINOIS TECH | Stuart School of Business

Plan of Action

Impact of AI on Employees, Management, and Jobs:

With the employment of drones, management will have to make significant changes to the workforce both in the systems used as well as the people. The primary mode of delivery through the use of its postal vehicle service which once shifted towards drone technology will replace those workers with drone technicians that won't require the round-the-clock labor. Drivers would be required to develop more software-based skills such as monitoring the drone software per each zip code and will be assisted by a team of drone technicians to help understand how a drone operates, its mechanics and software. This would ease the relative stress on veterans as they would focus more on the maintenance of the package and overall drone checks before they are deployed. As per credible sources[1], more than half of the workforce will be eligible for retirement by the next decade which would force the youth to build on their programming knowledge.

Supervisors for each central hub will be assisted with a CDOM (chief drone operations manager) that will help guide their understanding of overseeing fleets, ensuring that all drone regularly undergo maintenance and carry out the package delivery efficiently. Additionally, they would be managing a data analytics team who will be responsible for drone diagnostics in particular areas based on weather, temperature, altitude, aerodynamics, etc and analysis of customer satisfaction and popularity through this choice of package transportation per area. Management will also see appropriate training in courses for drone maintenance and package diagnostics to better understand the required software for each drone fleet which will involve all trainees. They will premium access to websites such as Coursera and Udemy for software-based knowledge and attend the drone training course by Illinois Aviation academy that is one of the top ranked schools for drone technology as well as regulations of laws applied per state.

Human-Machine Collaboration Opportunities

The workforce would be divided into 2 categories:

- Drone and package maintenance
- Unmanned Flight Diagnostics and customer support

Drone and package maintenance:

[1] https://www.uspsoig.gov/reports/white-papers/examining-trends-postal-services-workforce-composition#:~:text=The%20Postal%20Service%20is%20facing,retire%20within%20the%20next%20decade.

ILLINOIS TECH | Stuart School of Business

This team would comprise of technicians and much of the older workforce. Drone maintenance is highly prioritized given that its main mode of delivery and having been overseen by the technicians will ensure that packages be delivered as per the schedule. Each of the workers will be divided into groups that will handle particular fleets for each zip code with last minute drone maintenance and evaluation. Separate workers will ensure that all contents of the package are in the same condition as when taken for delivery with an inventory for such packages kept in database that is monitored continuously. When packages are delivered the customer can mark off contents that are present or missing so that the management can ensure the contents haven't been mishandled.

Unmanned Flight Diagnostics and customer support:

This team would comprise drone pilots, drone assistance team and maintenance, data analysts and human resources. Drone pilots will be in charge of reporting the condition of the drone and package at periodic intervals and ensuring the drone is maintaining a viable path to its destination which will be backed up by the assistant team monitoring the status of the drone and geographic conditions. The data analysts would be in charge of collecting and analyzing data in a server of the branch for diagnostics. This will also include ensuring the customer has successfully received the package and whether they choose to continue with the service should the package meet their expectations. Their analysis would be key understanding the surrounding topography and create more efficient flight paths for areas with similar conditions. Customer feedback is the most effective means of collecting data about conditions of the package as well as customer satisfaction. Any queries regarding current conditions of the package will be updated by customer support directly to a portal where customers can verify the trajectory of the package as well as its current condition in real time. Due to a major number of older workers most of them will be part of HR team to guide them through the process. If required, there will be an additional team on standby in case of hardware or software issues that will immediately notify the customer of the situation and have a secondary drone in place to complete the delivery.

<u>Ethical Considerations & Mitigation Strategies</u>

Job Displacement: While drone technology allows delivery drivers to become drone pilots, those who are unable to make the shift, particularly those who lack the technical abilities necessary for drone piloting and maintenance, face the possibility of job loss. This transition from conventional delivery positions to more tech-focused professions may leave some workers behind, particularly those nearing retirement or unfamiliar with new technology.

Privacy Violations: As drones become more common in mailing services, they may mistakenly catch footage of private property or persons, presenting serious privacy problems. Although their primary responsibility is package delivery, the capacity to capture or transmit video might result in unintentional breaches of human privacy, making this an ethical issue that businesses must handle.

ILLINOIS TECH | Stuart School of Business

Security Risks: The dependence on drones exposes vulnerabilities to cyberattacks. Hackers might possibly take control of drones to intercept goods or compromise personal information, raising security risks that must be addressed proactively.

To address these ethical issues, USPS might focus on reskilling programs that provide delivery drivers with the required training to become drone pilots and technicians, therefore mitigating the impact of job displacement. In terms of privacy, businesses should implement stringent data protection policies that limit data acquisition to only necessary information. To mitigate security threats, modern cybersecurity solutions, such as encryption and secure flight control, should be used to prevent hacking and assure safe, uninterrupted delivery. This method aids in balancing technical advancement with ethical issues.

Rubric

Your Module 4 assignment will be reviewed according to the following rubric:

ILLINOIS TECH
Stuart School of Business

ILLINOIS TECH | Stuart School of Business

	Criteria not met 0 points	Criteria met **4.0 points**	Good **4.5 points**	Exceptional **5.0 points**
Question 1: *Executive Summary*	No submission. OR The executive summary fails to capture the essence of the roadmap.	The executive summary captures the **basic** elements but lacks full effectiveness.	The executive summary is clear and **effectively** communicates the essence of the roadmap.	The executive summary is **exceptionally** clear, concise, and effectively captures the full value of the roadmap.
Question 2: *Current State*	No submission or the current state section fails to provide an accurate analysis of the organization.	The current state section captures **basic** details but lacks depth or clarity.	The current state section provides a **well-rounded and clear** analysis of the organization.	The current state section provides an **exceptionally detailed and insightful** analysis of the organization.
Question 3: *Proposed Initiative*	No submission or the proposed initiative lacks feasibility or relevance to the organization's needs.	The proposed initiative is **relevant** but may lack full feasibility or alignment with the organization's goals.	The proposed initiative is **well-aligned** with the organization's goals and is feasible.	The proposed initiative is **highly aligned**, feasible, and demonstrates deep strategic insight.

ILLINOIS TECH
Stuart School of Business

ILLINOIS TECH | Stuart School of Business

Question 4: *Plan of Action & Criteria for Success*	No submission or the plan of action lacks clarity, logical structure, or feasibility.	The plan of action is **clear** but may lack detail or full feasibility.	The plan of action is **well-structured, clear, and feasible** with solid criteria for success.	The plan of action is **exceptionally clear, well-structured, highly feasible**, and includes strong criteria for success.
Question 5: *Integration and Coherence*	No submission or the document fails to integrate sections effectively, lacking logical flow.	The document **attempts** to integrate the sections but lacks full coherence.	The document **integrates** the sections with good coherence and logical flow.	The document **integrates all sections seamlessly** with exceptional coherence and logical flow.
Question 6: *Structure and logic of writing:*	No submission. OR The response is unstructured, unclear, and lacks succinctness.	The response is **logically structured**, clear enough to comprehend, and mostly succinct.	The response is **well structured** in terms of logic, clarity, and succinctness.	The response is **exceptionally well structured** in terms of logic, clarity and succinctness.
Total points (example): **5+5+5+5+4.5+5 = 29.5** **Grade: 4.91 (5 !)**				